



Large-scale inventory in natural forests with mobile LiDAR point clouds

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ARTICLE INFO

Keywords:

LiDAR
Forestry
Deep learning
Forest inventory
Stem mapping
Diameter at breast height

ABSTRACT

Large-scale forest inventory at the individual tree level is critical for natural resource management decision making. Terrestrial Laser Scanning (TLS) has been used for individual tree level inventory at plot scale. However, due to the inflexibility of TLS and the complex scene of natural forests, it is still challenging to localize and measure every tree at large scale. In this paper, we present a framework to conduct large-scale natural forest inventory at the individual tree level by taking advantage of deep learning models and Mobile Laser Scanning (MLS) systems. First, a deep learning model, ForestSPG, was developed to perform large-scale semantic segmentation on MLS LiDAR data in natural forests. Then, the forest segmentation results were used for individual stem mapping. Finally, Diameter at Breast Height (DBH) was measured for each individual stem. Two natural forests mapped with backpack and Unmanned Aerial Vehicle (UAV) LiDAR systems were tested. The results showed that the proposed ForestSPG is able to segment large-scale forest LiDAR data into multiple ecologically meaningful classes. The proposed framework was able to localize and measure all 5838 stems at individual tree level in a 20 ha natural forest in less than 20 min using UAV LiDAR. DBH measurement results on trees' DBH larger than 38.1 cm (15 in) showed that backpack LiDAR was able to achieve 1.82 cm of Root Mean Square Error (RMSE) and UAV LiDAR was able to achieve 3.13 cm of RMSE. The proposed framework can not only segment complex forest components with LiDAR data from different platforms but also demonstrate good performance on stem mapping and DBH measurement. Our research provides an automatic and scalable solution for large-scale natural forest inventory at individual tree level, which can be the basis for large-scale estimation of wood volume and biomass.

1. Introduction

Forest inventory, the process of systematically measuring and assessing trees' health, composition, and distribution within a forest, is essential for effective forest management and conservation. Forest inventory data are used to inform policy decisions, monitor changes in forest ecosystems over time, and plan sustainable management practices (Romijn et al., 2015). Traditional forestry inventory methods are manual and rely on the expertise of forest professionals. This kind of method is often laborious and time-consuming, which can result in outdated or incomplete forest inventory data. Therefore, it hinders effective management and conservation efforts. As forest ecosystems continue to face pressures from climate change, deforestation, and other human activities, the availability of large-scale and accurate forest inventory data has never been more critical.

Light Detection and Ranging (LiDAR) technique has been proven effective for forest inventory (Liang et al., 2012; Newnham et al., 2015), offering numerous advantages over traditional methods (Liang et al., 2016). One significant advantage is that LiDAR can obtain forest 3D point clouds quickly and accurately with non-destructive scanning (Seidel et al., 2011). Terrestrial laser scanning (TLS), mounting LiDAR sensors on a tripod with a fixed position, is capable of capturing very fine 3D point clouds of forest structures with high precision (Calders et al., 2020). Therefore, mapping stems and measuring Diameter at Breast Height (DBH) are possible with TLS LiDAR data (Liang et al., 2016). To cover larger areas, however, TLS requires multiple scans from different fixed locations, which is not flexible and scalable (Wilkes et al., 2017). Due to the lack of Global Navigation Satellite System (GNSS) information associated with TLS, these scans need to be registered manually in order to create a comprehensive and precise 3D point cloud (Guan et al.,

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<https://doi.org/10.1016/j.srs.2024.100168>

Received 31 July 2024; Received in revised form 16 September 2024; Accepted 28 September 2024

Available online 9 October 2024

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2020). Even with GNSS information, manual registration would still be necessary due to differences in scan-to-scan orientation (Hasheminasab et al., 2023). The registration process is time-consuming and challenging, particularly in areas with limited accessibility or rough terrain (Kelbe et al., 2016).

In contrast, Mobile Laser Scanning (MLS), mounting LiDAR scanner on an Unmanned Aerial Vehicle (UAV) or a backpack, strikes a balance between area cover and point density, offering greater flexibility and scalability (Lin et al., 2022). These systems, especially UAV LiDAR systems, can be relatively easy to deploy in challenging terrain, allowing for more efficient and accurate forest LiDAR data acquisition, and collecting high point density data over a large area in a relatively short time. Additionally, UAV and Backpack LiDAR systems often integrate GNSS/Inertial Navigation System (INS) technology, enabling more accurate and automatic georeferencing and data integration without the need for manual registration (Lin et al., 2021b).

Backpack LiDAR mapping systems have been explored for forest inventory in the last decade and achieved significant progress. Kukko (2013) and Liang et al. (2014) first developed a backpack LiDAR system and conducted the first trial for forestry research. Subsequently, an increasing number of forest studies have used backpack LiDAR System (Ko et al., 2021; Holmgren et al., 2017; Su et al., 2021; Hyypä et al., 2020). Ko et al. (2021) found that using a backpack LiDAR system for stem mapping is four times faster than conventional methods. Holmgren et al. (2017) conducted DBH measurements for four plots of Norway Spruce with a 20 m radius. Su et al. (2021) developed a backpack LiDAR system and measured DBH from six sites of forests with the size of 30 × 30 m in different forest types, the results show that data derived from the backpack LiDAR system is equivalent to TLS data in terms of precision for DBH measurements.

Meanwhile, UAV LiDAR mapping systems have also been used for forest inventory (Jaakkola et al., 2010; Wallace et al., 2012; Dalla Corte et al., 2020; Lin et al., 2022). Jaakkola et al. (2010) first integrated UAV and LiDAR sensor with GNSS and Inertial Measurement Unit (IMU) units and showed its potential for forest inventory. Wallace et al. (2012) developed a low-cost UAV LiDAR system and proved that it has repeatability in the measurement of individual tree height and canopy width. Dalla Corte et al. (2020) selected 63 trees from a tree plantation in Brazil and measured their DBH and height with UAV LiDAR, which is beneficial to automatic tree plantation inventory. Lin et al. (2022) comprehensively investigated multi-temporal LiDAR data from both backpack and UAV laser scanning in a red oak plantation, and they found that both backpack and UAV LiDAR data are capable of individual tree localization.

Despite advancements in mobile LiDAR for forest inventory, large-scale natural forest stem mapping and measurement with billions of points are still facing processing efficiency and generalizability issues. First, previous methods relied on calculating linear or cylindrical point-wise geometric features (Chen et al., 2018), which can be time-consuming due to recursive searches (Chisholm et al. 2013; Lin et al. 2021a), especially for large amount of points. Additionally, geometric features lack semantic information, leading to missed detection of irregularly shaped stems in complex environments. Recent approaches use semantic segmentation with advanced machine learning methods to address these issues, showing its effectiveness across various LiDAR platforms, though it had limited number of points as input Krienski et al. (2021). While several deep learning methods have been developed for large-scale point cloud analysis (Hu et al., 2020; Landrieu and Simonovsky, 2018), their effectiveness in complex natural forests remains unproven. Finally, different LiDAR platforms have varying data properties due to differences in sensor position, distance, and reflection angle, leading to challenges in applying methods developed for one platform to data from another.

In this paper, we present a fully automated and scalable framework designed for conducting forest inventories in large-scale, complex natural forests using backpack and UAV Mobile LiDAR Mapping Systems.

Our approach integrates deep learning-based semantic segmentation, stem mapping, and diameter measurements. Notably, the framework achieves automatic stem mapping and diameter measurement across large-scale forests without the need for intricate parameter tuning. This adaptable solution for forest inventory leverages mobile LiDAR data from both backpack and UAV systems, addressing the growing demand for detailed, accurate forest data at scale through remote sensing technology.

2. Data

2.1. LiDAR systems

Two in-house developed UAV LiDAR systems (UAV1 and UAV2), and one commercial backpack LiDAR system (BP) were used in this study. Each in-house developed UAV LiDAR system is equipped with a multi-beam spinning laser scanner, camera, and GNSS/INS unit for direct georeferencing. System calibration is conducted using the in-situ calibration procedure proposed by Ravi et al. (2018). UAV1, a DJI M600 drone, is equipped with a Velodyne VLP-32C laser scanner and georeferenced through an Applanix APX-15v3 position and orientation GNSS/INS unit. UAV2, an AltaX drone, is equipped with an Ouster OS2-64 LiDAR scanner and Applanix APX-15v3 GNSS/INS unit. BP, the commercial backpack LiDAR system from Emesent named Hovermap ST, is equipped with a Velodyne 16-beam LiDAR scanner but without a GNSS/INS unit.

2.2. Study area

Our research was conducted in Martell Forest, a temperate forest in West Lafayette, Indiana, USA (Fig. 1). Within Martell Forest, we selected two stands of natural forests. The first stand is Plot 4D, a natural forest with an area of roughly 8 ha and relatively flat terrain. It contains at least 20 species dominated by oak (*Quercus* spp.) and maple (*Acer* spp.), the largest DBH is around 89 cm (35 in), and the maximum tree height is around 40 m. The second stand is Plot 4B, another natural forest in the south of Plot 4D with an area of roughly 20 ha. Plot 4B received a prescribed burn in Fall 2022 and a selective harvest within the last 5 years. This leads to fewer mature canopy trees less mid-story vegetation but complex understory vegetation. The composition of tree species and sizes of Plot 4B is similar to Plot 4D.

2.3. Data acquisition

LiDAR point cloud. Three LiDAR point clouds from the above two natural forests were collected in the leaf-off season as follows: Plot4D_BP, Plot4D_UAV1, and Plot4B_UAV2. Fig. 1 shows the spatial coverage of these three point clouds and Table 1 shows attributes of these point clouds. Plot4D_BP data were acquired on March 7, 2023, which took 35 min to scan an area of 2.25 ha. The LiDAR point cloud data for Plot4D_UAV1 was collected on March 3, 2021, covering an area of 7.2 out of 8 ha. The terrain for Plot 4D is relatively flat, so flight parameters were a fixed altitude of 38 m above ground, a speed of 3.5 ms⁻¹, and a lateral line spacing of 11 m. It took 40 min to cover all 7.2 ha of Plot 4D. For Plot 4B with larger areas and more sloping terrain, it is hard to capture consistent point density data with a fixed flying height. To solve this problem, we kept a fixed height difference between the UAV and the terrain of 62 m. Flight speed was 3.8 ms⁻¹ with a lateral line spacing of 8 m. Flight time was 60 minutes to complete the data acquisition over Plot 4B on April 19, 2023. While UAV LiDAR point clouds have been georeferenced due to UAVs' GNSS/INS unit, the BP LiDAR point cloud was not georeferenced due to its lack of GNSS information. We manually selected four pairs of distinctive points to calculate a transform matrix to perform coarse registration, and then used the Iterative Closest Point (ICP) algorithm (Besl and McKay, 1992) to perform fine registration.

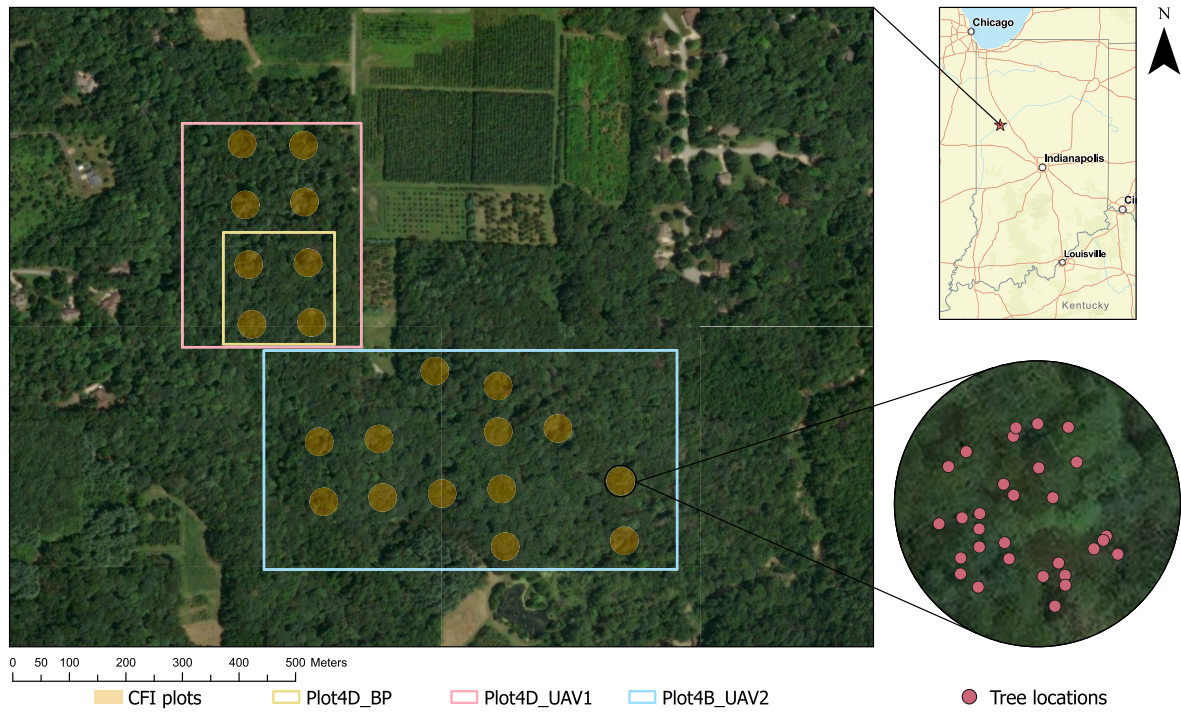


Fig. 1. The map of the study area. Two natural forests, Plot 4D (pink box) and Plot 4B (blue box) are located in Martell Forest in Indiana, USA (the top-right figure). Plot4D_UAV1 (pink box) and Plot4B_UAV2 (blue box) were scanned by two different UAV LiDAR systems. Plot4D_BP (yellow box), a part of Plot 4D was scanned by a backpack LiDAR system. The bottom-right figure shows a close view of one of the CFI plots, each point represents an individual stem location. Plot4D_UAV1 and Plot4B_UAV2 contain 8 and 13 CFI plots, respectively. Plot4D_BP contains 4 CFI plots that are shared with Plot4D_UAV1.

Table 1

Attributes of the three sets of LiDAR point cloud.

Data	Sites	Date	Platforms	Areas (ha)	Data acquisition time (min)	Num of points (million)	Point density (points/m ²)	Num of CFI plots
Plot4D_BP	Plot 4D	03/07/2023	backpack	2.25 ha	35	112	4894	4
Plot4D_UAV1	Plot 4D	03/03/2021	UAV1	7.2 ha	40	184	2486	8
Plot4B_UAV2	Plot 4B	04/19/2023	UAV2	20 ha	60	436	2292	13

Continuous Forest Inventory (CFI) data. In these two areas, CFI data were collected for evaluation based on the U.S. Forest Service Forest Inventory and Analysis (FIA) standard during March 2022. Plot 4D contains 8 CFI plots and Plot 4B contains 13 CFI plots (Fig. 1). For each CFI plot, trees over 12.7 cm (5 in) in DBH located within 16 m plot radius, were identified and measured. For each of these target trees, azimuth and distance to the plot center and its DBH were measured. The global coordinate of each tree in the plot was determined by using the azimuth and distance measures to calculate latitude and departure from the plot center for each tree and then adding these values to the plot center coordinates.

The standard practice of using GPS to measure CFI plot centers does not provide the needed accuracy for this study since both navigation-grade devices and survey-grade GNSS receivers often produce coordinate measurements containing multi-meter error when used under the forest canopy (Wing, 2008; Edson and Wing, 2012; Naesset, 2001). To obtain CFI plot center coordinates with sufficient accuracy to confidently pair with the corresponding ground-based measurements, this study follows the procedure introduced by Carpenter et al. (2023): cost-effective targets were deployed at every plot center, UAV imagery is then captured over the area of interest during leaf-off conditions, and then plot center coordinates were manually extracted from the ortho imagery.

2.4. Data annotation

A well-annotated dataset was created for training and evaluation. Our goal is to perform a large-scale detailed forest inventory, so we need to carefully semantically classify all points from the LiDAR point cloud to extract useful information. To evaluate the segmentation results and provide accurate supervised information for model training, an accurately annotated dataset is essential. We first defined the compositions of forests in 5 classes: ground, debris, trunk, crown, and shrub. One should note that we define trunk and crown according to crown base height (CBH, the vertical distance between the ground and the base of the live crown). The portions lower than CBH are defined as trunks and higher than CBH are defined as crowns (Popescu and Zhao, 2008). Fig. 2 illustrates the class definition. To make sure that the dataset is accurate enough for model training and evaluation, all points have been annotated and cross-examined by two individuals. The entire LiDAR point cloud dataset is split into training (including validation) and testing sets. The training set only contains the data of Plot4D_UAV1 that is not covered by CFI areas. To evaluate model accuracy and generalization ability, the testing set contains areas that cover all the CFI data in Plot4D_BP and Plot4D_UAV1. Plot4B_UAV2 data that are covered by CFI plots are only used to evaluate the effectiveness of stem mapping and DBH measurements.

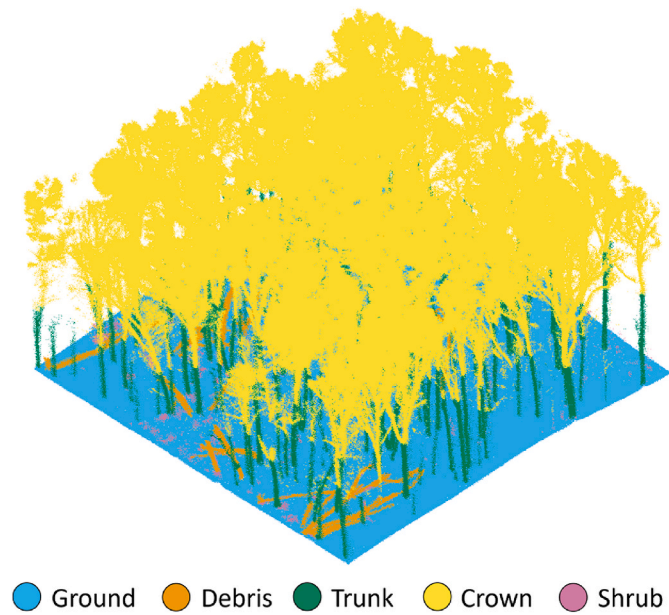


Fig. 2. Illustration of class definition. A 50×50 m natural forest point cloud is colored by classes and they are ground, debris, trunk, crown, and shrub.

3. Methods

The fully automated framework we developed for large-scale natural forest inventory consists of three steps (Fig. 3): (1) Semantic segmentation of natural forest with ForestSPG; (2) stem mapping from results of semantic segmentation; and (3) DBH measurements of individual stems. A natural forest is first segmented into different classes including

ground, debris, trunk, crown and shrub. Then, the segmented tree trunks are further clustered into individual trunks. Finally, DBH is measured by using circle fitting.

3.1. ForestSPG

In natural forests, the environment is characterized by a rich tapestry of trees with diverse sizes and shapes, creating a complex landscape. Despite this diversity, forests exhibit homogeneous characteristics when analyzed within specific categories. For instance, ground points typically form planar features, while stem points present linear or cylindrical structures. This observation suggests that complex forest environments can be effectively deconstructed into simpler geometric primitives using over-segmentation techniques. These primitives, in turn, can be further classified into distinct ecological components by exploring their spatial and semantic relationships.

Building on this foundational insight, we introduce ForestSPG (Forest Superpoint Graph), an advanced deep learning framework built upon SPG (Landrieu and Simonovsky, 2018), designed to enhance semantic segmentation of forest point clouds for detailed inventory at the individual tree level. ForestSPG operates in two key stages: over-segmentation followed by semantic learning. In the first stage, ForestSPG decomposes the point cloud data into appropriately sized superpoints, which serve as the basis for classification. In the second stage, a graph neural network is employed to refine and learn the semantic information from these superpoints, thereby enabling accurate and comprehensive understanding of the forest's ecological components.

3.1.1. Over segmentation

The objective of this module is to partition the point cloud into multiple superpoints, with each of which contains a single geometric

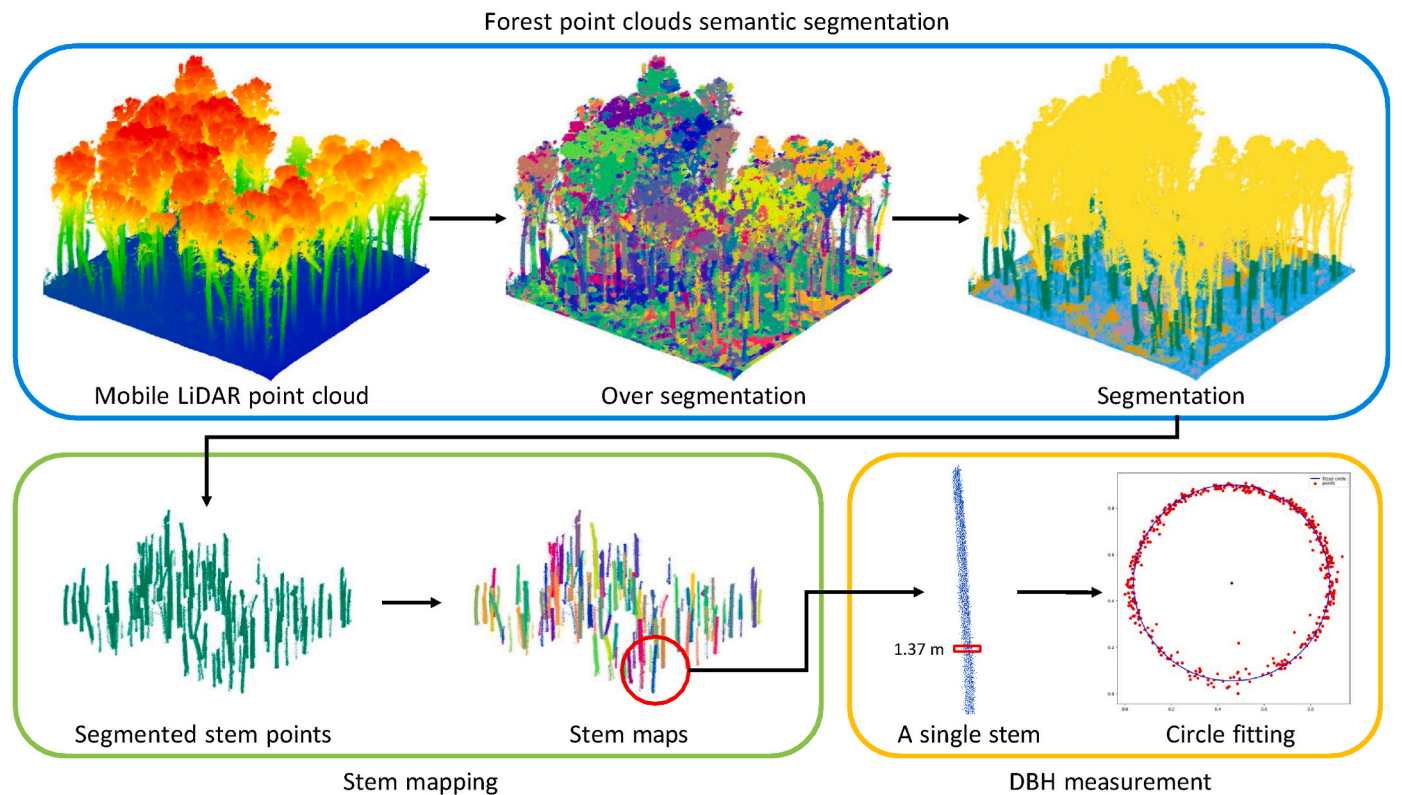


Fig. 3. The proposed framework for large-scale natural forest inventory with mobile LiDAR point clouds. Semantic segmentation is first performed in natural forest LiDAR point clouds (blue box) with ForestSPG. Then the segmented stem points are further clustered into individual stems (green box). For each clustered stem, a 10 cm cross-section at 1.37 m ($4\frac{1}{2}$) feet from the ground is cropped and the circle fitting algorithm is performed to evaluate the DBH (yellow box).

feature. We found that a superpoint generated by the original SPG sometimes contains two or more semantic components in a forest scene, causing under-segmentation. To resolve this issue, we improved the process by employing an over segmentation algorithm by Papon et al. (2013) to generate almost uniformly sized superpoints, allowing each superpoint only contains one semantic component. It has been proven that both radiometric and geometric features are helpful in separating different components of trees (Shao et al., 2023), so we also include both radiometric and geometric information into feature calculation to achieve over segmentation.

We first voxelize the input point cloud with a size of 5 cm for efficient processing. Then, several 3D descriptors are calculated for each point including linearity, curvature, and surface normal based on Demantké et al. (2012). After that, we uniformly select a set of seed points so that they are spatially evenly distributed throughout the point cloud. Starting from each seed point, we search the closest point among its 26 neighbors in an octree structure by calculating distance D defined below:

$$D = \sqrt{w_r D_r^2 + \frac{w_s D_s^2}{3R_{seed}^2} + w_g D_g^2} \quad (1)$$

where D_r , D_s , and D_g represent radiometric, spatial extent, and geometric Euclidean distances. R_{seed} represents the interval distance of each seed point. D_s is spatial distance normalized by the R_{seed} . It is worth noting that D_r is the value difference between intensity values, and D_g is the combination of linearity, curvature and surface normal. According to this calculation of D , we gradually set newly added points as new centers to grow the superpoint until it meets the sphere boundary (i.e., R_{seed}). We manually set R_{seed} to 50 cm, and set w_r , w_s , and w_g to 0.1, 0.5, and 1 respectively after trial and error.

Once the over segmentation is done, a superpoint graph is generated by constructing a 3D Voronoi adjacency graph among the partitioned point cloud, which can be represented as: $G_{SP} = (V_{SP}, E_{SP})$. V_{SP} represents superpoints and E_{SP} represents superedges.

3.1.2. Semantic learning

In the learning process, each over-segmented superpoint will be assigned a semantic label to achieve semantic segmentation. Each superpoint is first fed into a PointNet (Qi et al., 2017), a well-known neural network for point cloud classification and segmentation, to obtain an embedding vector. The PointNet input contains 3D coordinates and geometric features that were derived from the over segmentation step. The output of the PointNet, superpoint embedding, can be seen as a low-dimensional superpoint descriptor that is ready to be the input for a graph convolution. Next, a graph convolution network is used to classify superpoints. The graph convolution network is based on Gated Graph Neural Networks and Edge-Conditioned Convolutions (ECC) (Li et al., 2015; Simonovsky and Komodakis, 2017). Each superpoint embedding is fed into a gated recurrent unit (GRU) by message passing along with superedges. Each superedge weight is computed by a multi-layer perception. Finally, an ECC is introduced to conduct graph convolution to classify every superpoint. The output of ECC for each superpoint contains prediction scores for each class, and the class with the highest score will be the prediction.

3.1.3. Data augmentation and model training

Data augmentation. Data augmentation has been widely used for point cloud understanding tasks such as point cloud classification, semantic segmentation, and object detection (Chen et al., 2020). A simple but effective way is to directly apply random scaling, rotation, and translation to single objects. However, such random rotation and translation would not fit our input which is a big area of forest instead of small objects (e.g., trees). Because there are no forests that grow downwards or sideways in the real world. To make data augmentation fit the real situation, we propose holistic forest data augmentation for forest point clouds. For every input point cloud, we perform global flipping by

orientation of horizontal and vertical (i.e., east-west and south-north). We also rotate tiles every 45 deg. In this way, 9 additional augmented tiles are generated.

Model training. large-scale forest point clouds are first tiled into 50×50 m patches. For each patch of point cloud, we perform holistic forest data augmentation. Then the augmented point clouds are fed into ForestSPG to perform semantic segmentation. As commonly done in deep learning, we interactively learn the ForestSPG parameters with stochastic gradient descent with random initialization. The initial learning rate is set to 0.01 and learning rate decay is applied during the model training with 0.8 decay rate. And it is automatically applied when the validation loss has not improved for 10 epochs. Finally, the model training would be stopped when the validation accuracy has not increased for 30 epochs.

The weighted cross entropy loss (Aurelio et al., 2019) is used for optimization of model parameters. It has been demonstrated to be effective in alleviating the class imbalance problem for point cloud segmentation (Han et al., 2021). We introduce this loss to forest point cloud segmentation and it is defined as:

$$L = \sum_{c=1}^M \frac{\prod_{i=1, i \neq c}^M \sqrt{N_i}}{\sum_{i=1}^M \sqrt{N_i}} y_c \log(\hat{y}_c) \quad (2)$$

where N_i is the total number of points in i^{th} class and M is the number of classes. y_c represents reference value of the class c and $\hat{y}_c$ represents the predicted value of the class c .

3.2. Stem mapping

In this study, we consider stem mapping as a stem detection task, which means conducting localization and counting by aggregating all trunk points into individual trunks. To achieve this, we use Connected Components Labeling (CCL), a simple yet efficient clustering algorithm (Rosenfeld and Pfaltz, 1966). Segmented stem points from the ForestSPG result are first voxelized using 3D grids by building an Octree. In the voxelized point cloud, all voxels that contain points are considered as foreground, while vacant voxels are denoted as background. Then, all foreground voxels are scanned for label assignment. In label assignment, if a voxel under scanning has no neighboring voxels with an assigned label, it is attributed a new label, thus initiating a new component. Alternatively, if the voxel's neighbors already owned a label, the voxel is allocated with the same label. If the voxel's neighbors have multiple labels, the voxel is assigned one of the labels, and the differing labels are recorded as equivalents and thereby forming part of the same component. Finally, all equivalent labels are unified resulting in individual stem clustering. Smaller grids make it easier to identify the gaps between trunks and separate them while increasing computation requirements. To make a compromise between the speed and efficiency, we suggest setting the grid size as the minimum distance between different adjacent trunks (such as 5 or 10 cm), which is small enough to separate the tree trunks. In addition, to remove some points that are misclassified from the ForestSPG result or noise, we compute all components' geometric features and remove non-linear features. It should be noted that irregular shape stems would still be kept as they are still linear features, while non-linear features here are some noise clusters.

3.3. DBH measurement

As one of the most classical attributes of forest inventory, DBH is often the primary information of concern. Due to the mobility of MLS LiDAR, the full shape of trunks can be scanned with limited omission. Therefore, least square circle fitting is used to accurately measure the DBH from a tree stem point cloud. For each detected trunk, we cut out a 10 cm cross-section at a height of 1.37 m (4.5 ft) above the ground (Magarik, 2021). Then this cross-section is projected onto the XY plane. Points (x_i, y_i) of the cross-section on the XY plane are first normalized to

the center by subtracting its centroid (x_c, y_c) . Then a circle with the center (a, b) and a radius R can be estimated through least squares adjustment, and the objective function can be expressed as follows:

$$F(a, b, R) = \sum_{i=1}^n (r_i - R)^2, \quad (3)$$

where $r_i = \sqrt{(x_i - a)^2 + (y_i - b)^2}$ denotes the distance between the observed data point (x_i, y_i) and the estimated circle center (a, b) . Finally, the optimal values for (a, b) and R are determined by minimizing $F(a, b, R)$ using the Levenberg-Marquardt optimization algorithm (Levenberg, 1944). Once the radius R is determined, the DBH can be established as 2R of the fitted circle.

3.4. Evaluation

We evaluate the effectiveness of our forest inventory method on the next three aspects: (a) semantic segmentation of forest point clouds, (b) stem mapping, and (c) DBH measurement.

3.4.1. Semantic segmentation

The assessment of semantic segmentation is seen as an evaluation of point-wise classification on the testing set, so we can use True Positive (TP), False Positive (FP), True Negative (TN), and False Negative (FN) by comparing classification results and reference. Then, the metrics, precision (indication of commission errors), recall (indication of omission errors), F1-Score, and Intersection over Union (IoU) are calculated based on Equations (4)–(7), respectively. In addition, Overall Accuracy (OA) as given in Equation (8) is also calculated for the segmentation evaluation.

$$Precision_{seg} = \frac{TP}{TP + FP} \quad (4)$$

$$Recall_{seg} = \frac{TP}{TP + FN} \quad (5)$$

$$F1 - Score_{seg} = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad (6)$$

$$IoU_{seg} = \frac{TP}{TP + FP + FN} \quad (7)$$

$$Overall Accuracy_{seg} = \frac{TP}{Number\ of\ points} \quad (8)$$

3.4.2. Stem mapping

Assessing stem localization accuracy is seen as a kind of object-wise (stem-wise) assessment. It does not only evaluate the number of detected stems but also the locations of stems. In our case, we set detected stems that overlap with the reference as correct detections denoted as TP, detected stems that don't overlap with the reference map as wrong detections denoted as FP, and stems on the reference map that are not detected denoted as FN. The threshold for overlapping is set to a 0.5 m radius, which means if there is a detected stem within 0.5m of a reference stem location, it is recognized as a TP. If there is no reference within 0.5m of a detected stem, then this stem is recognized as an FP. If there is no stem within 0.5 m of a reference stem location, then it is recognized as an FN. The reason we add 0.5 m as a threshold is to mitigate the shift between the reference measurement and the point cloud position, which may come from the tilted stems or GNSS offset.

Then, precision and recall for stem mapping can be computed as:

$$Precision_{mapping} = \frac{Correct\ detections}{All\ detections} = \frac{TP}{TP + FP} \quad (9)$$

$$Recall_{mapping} = \frac{Correct\ detections}{All\ ground\ truth} = \frac{TP}{TP + FN} \quad (10)$$

3.4.3. DBH measurement

The core metric for DBH accuracy is the difference between the digitally measured DBH and the reference. In this study, Mean Absolute Error (MAE) and Root Mean Square Error (RMSE) are used to evaluate the performance of DBH measurements (Armstrong and Collopy, 1992).

$$MAE_{DBH} = \frac{1}{N} \sum_{i=1}^N |\hat{y}_i - y_i| \quad (11)$$

$$RMSE_{DBH} = \sqrt{\frac{1}{N} \sum_{i=1}^N (\hat{y}_i - y_i)^2} \quad (12)$$

where N is the number of measured stems, y is reference DBH and $\hat{y}$ is predicted DBH.

It has been demonstrated that different data sources and sizes of tree DBH affect the measuring results (Dalla Corte et al., 2020). For each dataset, therefore, we divided stems into three groups (small, middle, and large) based on their DBH size classes, 12.7–25.4 cm (5–10 in), 25.4–38.1 cm (10–15 in) and 38.1–83.3 cm (15–34 in), to evaluate the performance separately.

4. Results

4.1. Forest point cloud semantic segmentation

The ForestSPG model was only trained on a part of Plot4D_UAV1 data, and evaluated on another part of Plot4D_UAV1 data and the entirety of Plot4D_BP. Table 2 shows quantitative results of the model on the testing set in terms of Precision, Recall, F1-Score, and IoU for both Plot4D_UAV1 and Plot4D_BP. The evaluation metrics are computed for all five classes - ground, debris, trunk, crown, and shrub. Sample visualization of segmentation results is presented in Fig. 4.

The ForestSPG model exhibited high Precision, Recall, F1-Score, and IoU values for the ground, trunk, and crown classes in the Plot4D_UAV1 data. The model was able to achieve 0.9 or higher metrics for ground and crown. Precision, recall, and F1-Score of trunk segmentation were in the range of 0.87 or higher, while IoU was 0.787. However, the model performance appeared to be significantly weaker for the debris and shrub classes. All metrics for debris and shrubs were lower than 0.4 except for the precision of debris, which was 0.698. Most misclassified debris and shrub points were recognized as ground. Despite these misclassifications, the model managed to maintain an overall accuracy of 98%.

Comparatively, the model performance on the Plot4D_BP presents a similar pattern. Although the BP data with higher point density has not been trained by our ForestSPG model, all metrics for ground, trunks, and crowns were above 0.9 and overall accuracy was also able to achieve

Table 2

The performance of the ForestSPG model on Plot4D_UAV1 LiDAR data in Plot4D from testing set.

		Precision	Recall	F1-Score	IoU
Plot4D_UAV1	Ground	0.982	0.995	0.988	0.976
	Debris	0.698	0.321	0.440	0.282
	Trunk	0.875	0.887	0.881	0.787
	Crown	0.991	0.994	0.992	0.984
	Shrub	0.476	0.119	0.190	0.105
	Overall Accuracy	98%			
Plot4D_BP	Ground	0.976	1.000	0.988	0.976
	Debris	0.217	0.166	0.188	0.104
	Trunk	0.906	0.996	0.949	0.903
	Crown	0.998	0.995	0.996	0.992
	Shrub	0.976	0.256	0.406	0.254
	Overall Accuracy	97%			

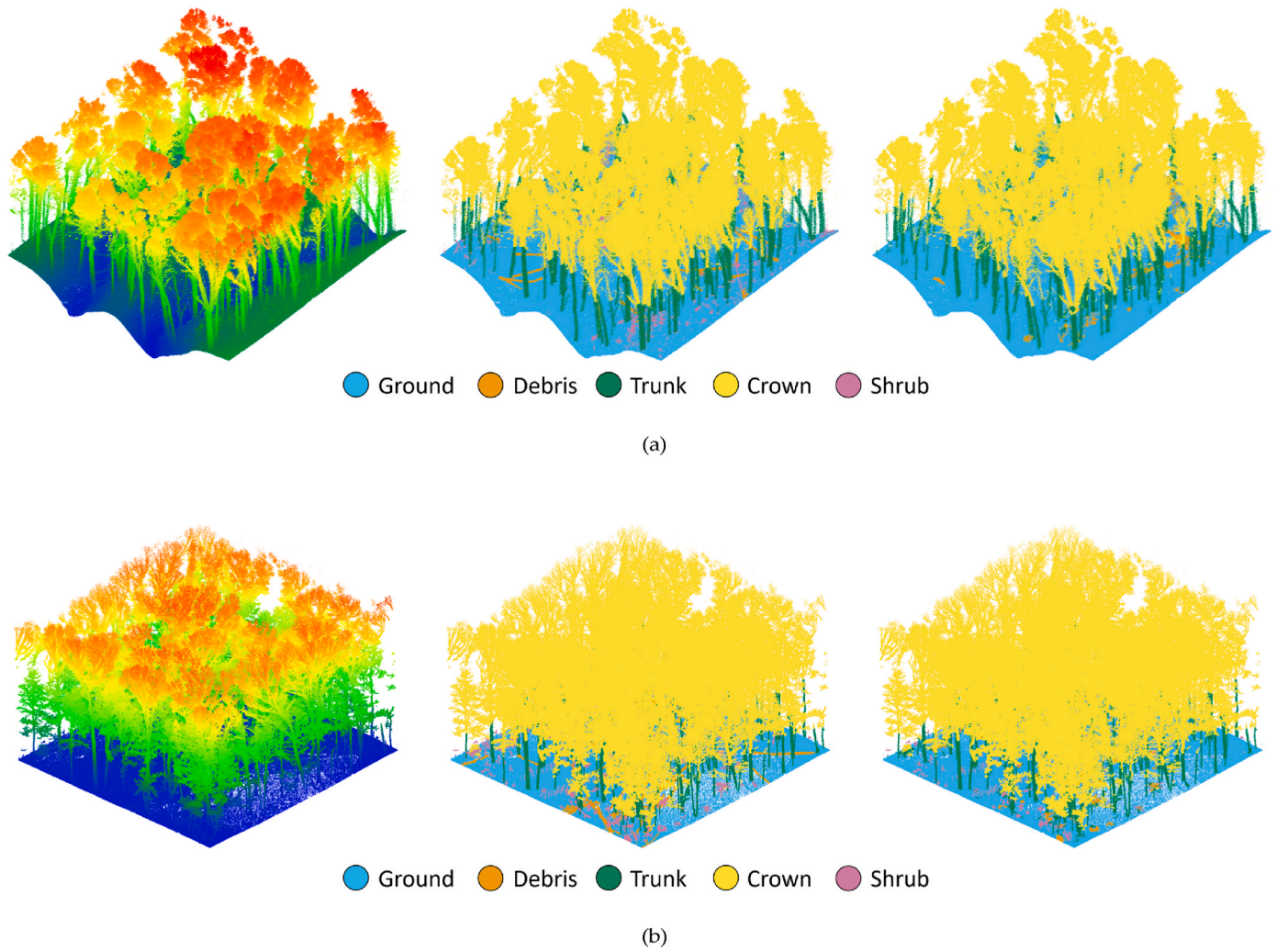


Fig. 4. Visualized semantic segmentation results for natural forests (a) a 50×50 m tile from Plot4D_UAV1 and (b) a 50×50 m tile from Plot4D_BP. From left to right: input point cloud colored by height, reference colored by classes, and prediction colored by classes.

97%. More surprisingly, the IoU value of BP LiDAR trunks segmentation was higher than its counterpart of UAV LiDAR. However, low values of metrics for the debris and shrub classes were also observed for the Plot4D_BP data, which posed a challenge.

4.2. Stem mapping

The stem mapping task, which involves the detection and localization of individual stems in the forest, was performed on three natural forest point clouds: Plot4D_BP, Plot4D_UAV1, and Plot4B_UAV2. The accuracy of the stem mapping on these datasets is summarized in Table 3.

For the Plot4D_BP dataset, out of 111 reference stems, 107 stems were successfully detected (TP) with no falsely identified stems (FP), leading to a precision of 1.00. However, it failed to detect 4 stems (FN), resulting in a recall of 0.96.

The stem mapping performance on the Plot4D_UAV1 dataset was

slightly lower. Out of the 207 reference stems, 198 were correctly detected while 5 stems were falsely identified, leading to a precision of 0.98. A total of 9 stems were not detected, which resulted in a recall of 0.96.

In the case of the Plot4B_UAV2 dataset, 187 out of 194 reference stems were correctly detected by the algorithm. This detection included 3 falsely identified stems, which corresponded to a precision of 0.98. There were 7 stems that were not detected, resulting in a recall score of 0.96.

4.3. DBH measurements

Quantitative DBH estimation results are presented in Table 4, and relationships between measured DBH and reference DBH from three different LiDAR platforms are shown in Fig. 5.

In general, Plot4D_BP achieved the best result DBH measurement with 1.64 cm of MAE and 2.29 cm of RMSE among all three data sources, Plot4D_UAV1 took second place with 3.50 cm of MAE and 4.31 cm of RMSE, and Plot4B_UAV2 obtained 3.68 cm of MAE and 4.83 cm of RMSE, which is similar with Plot4D_UAV1.

As the tree size increased, DBH measurement accuracy became better in both UAV LiDAR point clouds and the BP LiDAR point cloud (Fig. 5 and Table 4). For Plot4D_UAV1 and Plot4B_UAV2, groups of trees with small DBH obtained relatively high MAE (4.35 cm and 5.02 cm) and RMSE (5.08 cm of and 6.05 cm). Trees with large DBH obtained the best

Table 3
Performance of stem mapping for different data sources and study sites.

	TP	FP	FN	GT	Precision	Recall
Plot4D_BP	107	0	4	111	1.00	0.96
Plot4D_UAV1	198	5	9	207	0.98	0.96
Plot4B_UAV2	187	3	7	194	0.98	0.96

Table 4

MAE and RMSE of DBH measurements for three different data sources according to DBH ranges.

	DBH range	Num of trees	MAE	RMSE
Plot4D_BP	12.7–25.4 cm	53	1.65	2.37
	25.4–38.1 cm	15	2.08	2.97
	38.1–85.9 cm	43	1.45	1.82
Mean	12.7–85.9 cm	111	1.64	2.29
Plot4D_UAV1	12.7–25.4 cm	95	4.35	5.08
	25.4–38.1 cm	35	3.02	3.99
	38.1–85.9 cm	77	2.57	3.13
Mean	12.7–85.9 cm	207	3.50	4.31
Plot4B_UAV2	12.7–25.4 cm	80	5.02	6.05
	25.4–38.1 cm	39	3.23	4.29
	38.1–83.3 cm	75	2.11	2.86
Mean	12.7–85.9 cm	194	3.68	4.83

MAE (2.57 cm and 2.11 cm) and RMSE (3.13 cm and 2.86 cm). MAE and RMSE for groups of trees with medium DBH stayed in between small and large groups. The backpack LiDAR point cloud did not show such an intense pattern that DBH measuring had relative consistency in terms of DBH ranges. Overall, the backpack LiDAR demonstrated superior performance over UAV LiDAR in DBH measurements across all data sources. In addition, DBH measurements of larger DBH stems consistently outperformed their smaller counterparts in MAE and RMSE values, regardless of the data source.

5. Discussion

The fully automated and scalable framework for large-scale natural forest inventory we developed can be easily adapted to backpack or UAV LiDAR data. Mobile LiDAR Systems provide comparable LiDAR point clouds, in terms of point density, with TLS (Su et al., 2021). More importantly, MLS speed up data collection using GNSS/INS units at large scale without the need for manual registration. Such flexible and scaleable characteristics of MLS become the prerequisites for efficient large-scale forest inventory. Our experimental results prove that the backpack LiDAR that captures detailed and high-point density point clouds with our forest inventory framework is able to deliver 100% stem mapping accuracy and less than 1.64 cm of MAE for DBH measuring, which is extremely applicable for areas that are accessible by foot for on-the-ground surveying. For large areas that need detailed forest inventory information, our framework takes advantage of UAV LiDAR systems to obtain large-scale LiDAR point clouds in a short time to localize and measure every single tree with 98% stem mapping accuracy and less than 3.68 cm of MAE for DBH measuring.

Another advantage of our framework is that it overcomes the uncertainty and low frequency of traditional forest inventory methods. Forest inventory practices often require well-designed statistic-based sampling methods, which results in sampling errors of forest resources (Köhl and Magnussen, 2016). With our framework, the design of the sampling strategy is no longer needed as every stem is measured. In areas where inventory data is updated at low frequency, such as rugged hills, this solution can be used to obtain high-frequency forest inventory data. Such high-frequency forest inventory data is especially crucial for forest ecosystem monitoring, forest dynamics analysis, and carbon sequestration estimation.

5.1. Segmentation of large-scale complex natural forest

The deep learning-based method used in this research is able to segment large-scale complex natural forests into multiple ecologically meaningful semantic classes instead of simple binary classification of linear and non-linear points. Traditionally, extracting stem points relying on handcrafted linear or cylinder features can neither cope with complex natural forests nor extend to large-scale stem mapping. In addition, natural forests consist of multiple components (e.g. anisotropic shrubs, tangled trunks, and dead logs), so only extracting vertical cylinders cannot satisfy the natural forest inventory. In our framework, we presented ForestSPG, a deep learning method, for multi-class segmentation of mobile LiDAR point clouds in complex natural forests to avoid the above problems. Deep learning-based semantic segmentation makes it easy to classify different forest components and it also serves as the basis for other forestry applications, such as forest terrain analysis (ground segmentation), fire fuel estimation (debris and shrub segmentation), wood volume calculation (stem segmentation), and tree canopy cover (crowns segmentation).

Fig. 6 provides partial visualization of the model's input, prediction, and reference. It contains three 50 × 50 m tiles of point cloud data on flat and rugged terrains, respectively. As evident in the partial visualization, the segmentation model was able to separate ground and non-ground points clearly. Focusing on the components of trees, the model successfully segmented almost all trunks but had some minor difficulties in distinguishing tangled trunks. Although achieving good performance on ground and trunks, the model obtained unsatisfied results on debris and shrubs and most of them were misclassified as ground. In terms of terrain shape, the model met no difficulty in relatively flat terrain, while getting minor misclassifications of trunks in rugged areas. Overall, misclassified debris and shrubs didn't influence stem mapping accuracy, but performance of these categories could be improved in the future.

In terms of terrain shape, the model met no difficulty in relatively flat terrain, while getting minor misclassifications of trunks in rugged areas.

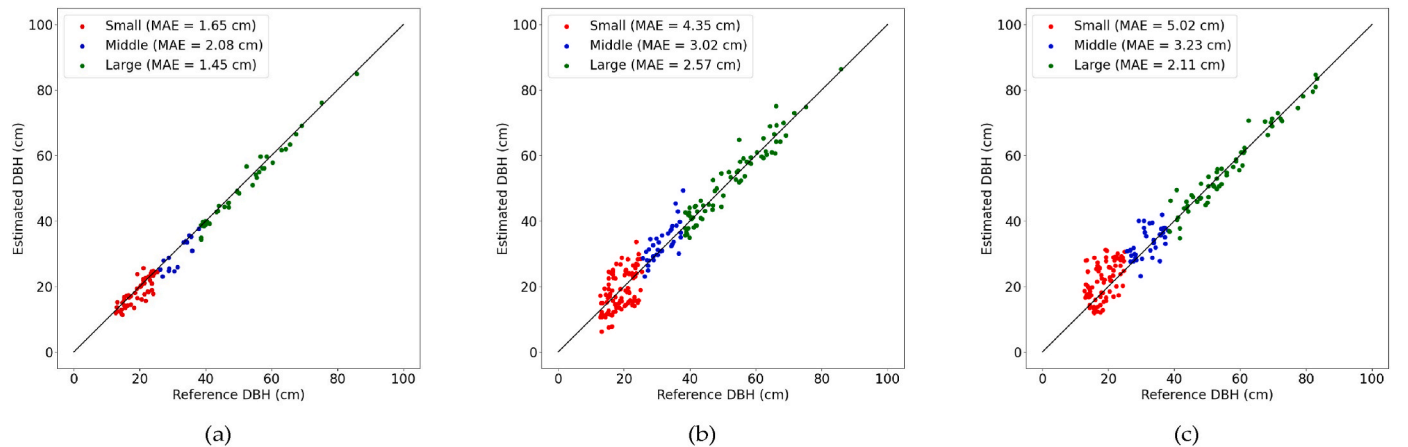


Fig. 5. Relationships between measured DBH and reference from (a) Plot4D_BP, (b) Plot4D_UAV1 and (c) Plot4D_UAV2.

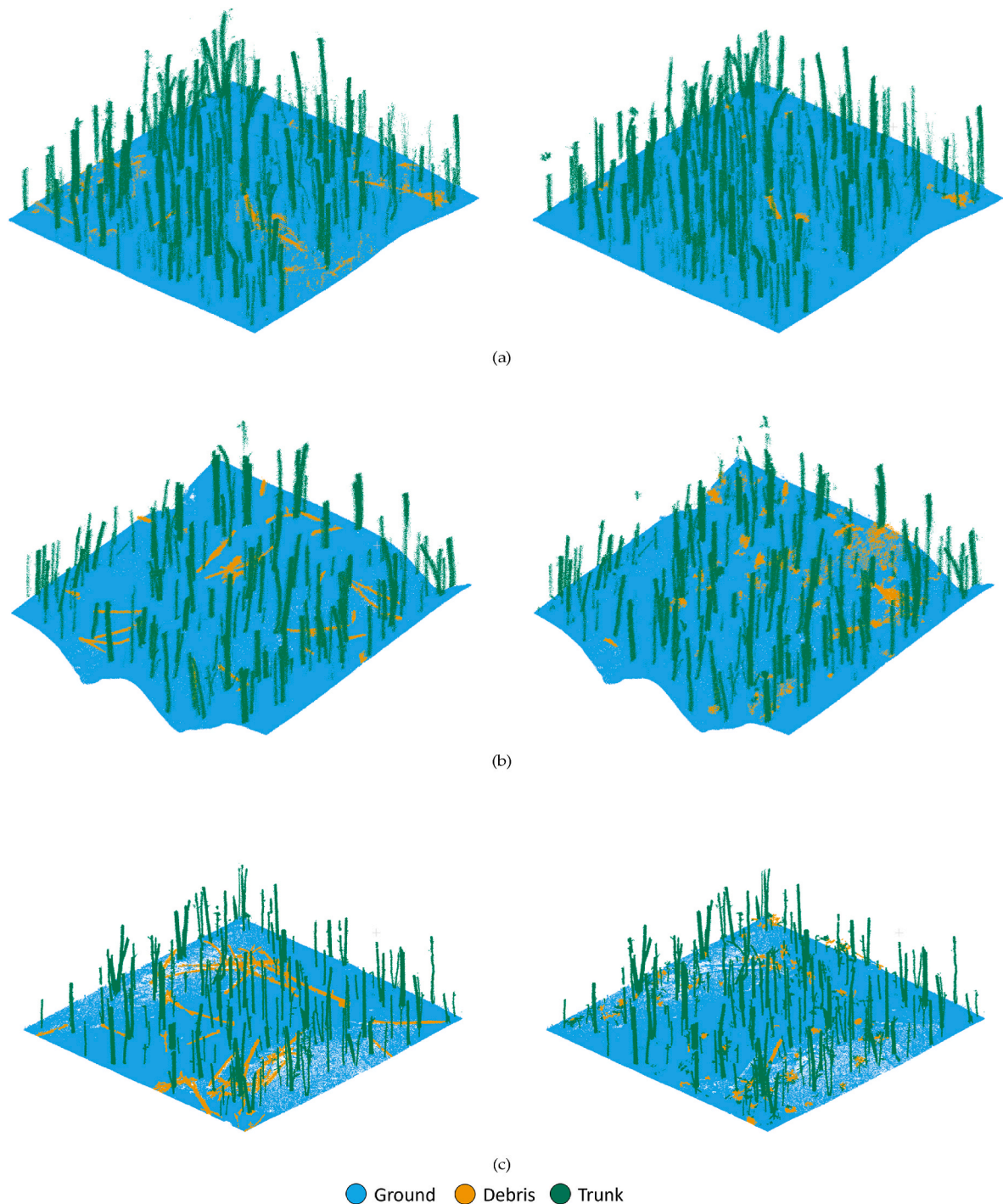


Fig. 6. Partial visualization of reference (left) and prediction (right) of natural forest LiDAR point cloud semantic segmentation results of 50×50 m tiles from (a) Plot4D_UAV1 with flat terrain, (b) Plot4D_UAV1 with rugged terrain, and (c) Plot4D_BP.

5.2. Generalization across LiDAR platforms

Another fact worth noting is that experimental results indicate that our ForestSPG has excellent generalizability across point cloud data from different LiDAR systems in natural forest scenes. The ForestSPG model was only trained on well-annotated Plot4D_UAV1 data, so it was not surprising that it could get favorable segmentation results on the testing set of Plot4D_UAV1. However, the experimental results of ForestSPG on unseen Plot4D_BP, LiDAR data from another platform, showed that its segmentation accuracy was on par with Plot4D_UAV1, which means our ForestSPG model is able to generalize from UAV to

backpack LiDAR data with higher point density for natural forest segmentation. Although we don't have reference data for Plot4B_UAV2, the stem mapping result of Plot 4B showed that tree counting and localization were derived from reliable segmentation results. This demonstrates the generalization of ForestSPG for natural forest segmentation across different UAV LiDAR platforms with different LiDAR sensors. Generalization of forest segmentation across data sources from different mobile LiDAR platforms allows the extension of the scalability and applicability of our framework. However, the generalizability across data from different LiDAR platforms was verified in temperate natural forests in the central United States, and its ability across different forest

types is yet to be verified, which will be one of our future research directions.

5.3. Efficiency of large-scale forest inventory

We also evaluated the efficiency of our framework on a 20 ha (50 acres) natural forest. The entire LiDAR data from UAV2, Plot4B_UAV2, were used for the efficiency evaluation of large-scale forest inventory (Fig. 7). As shown in Table 1, Plot4B_UAV2 contains 436 million points captured by our in-house UAV LiDAR system in 60 min. Then LiDAR point cloud data were processed on a server that consists of two Intel Ice Lake Silver CPUs, 8 NVIDIA RTX A5000 GPUs (24 GB memory for each GPU), and 256 GB RAM. Processing such 20 ha with 436 million points with our framework takes around 20 min on the server, at which 18 min

were used to parallel run point cloud over segmentation and superpoint graph construction on CPUs, 1 min for ForestSPG inference on GPU, and 1 min for stem mapping and diameter measuring. 436 million points were segmented resulting in a total of 5838 trees being localized and measured (Fig. 7b). It is worth noting that the over segmentation part can be run on personal desktop and training part can be done by using a GPU with 8 GB. After testing, one single RTX A5000 with 24 GB memory was able to run inference for 20000 m² forest at one time thanks to the low amount of parameters of superpoint graph. However, over segmentation and superpoint graph construction consumed the most processing time for the entire Plot 4B natural forest even running in parallel. Overall, for the entire Plot 4B, we finished forest inventory at individual tree level within 80 min including both LiDAR data collection and processing. Most of the time was dedicated to data collection.

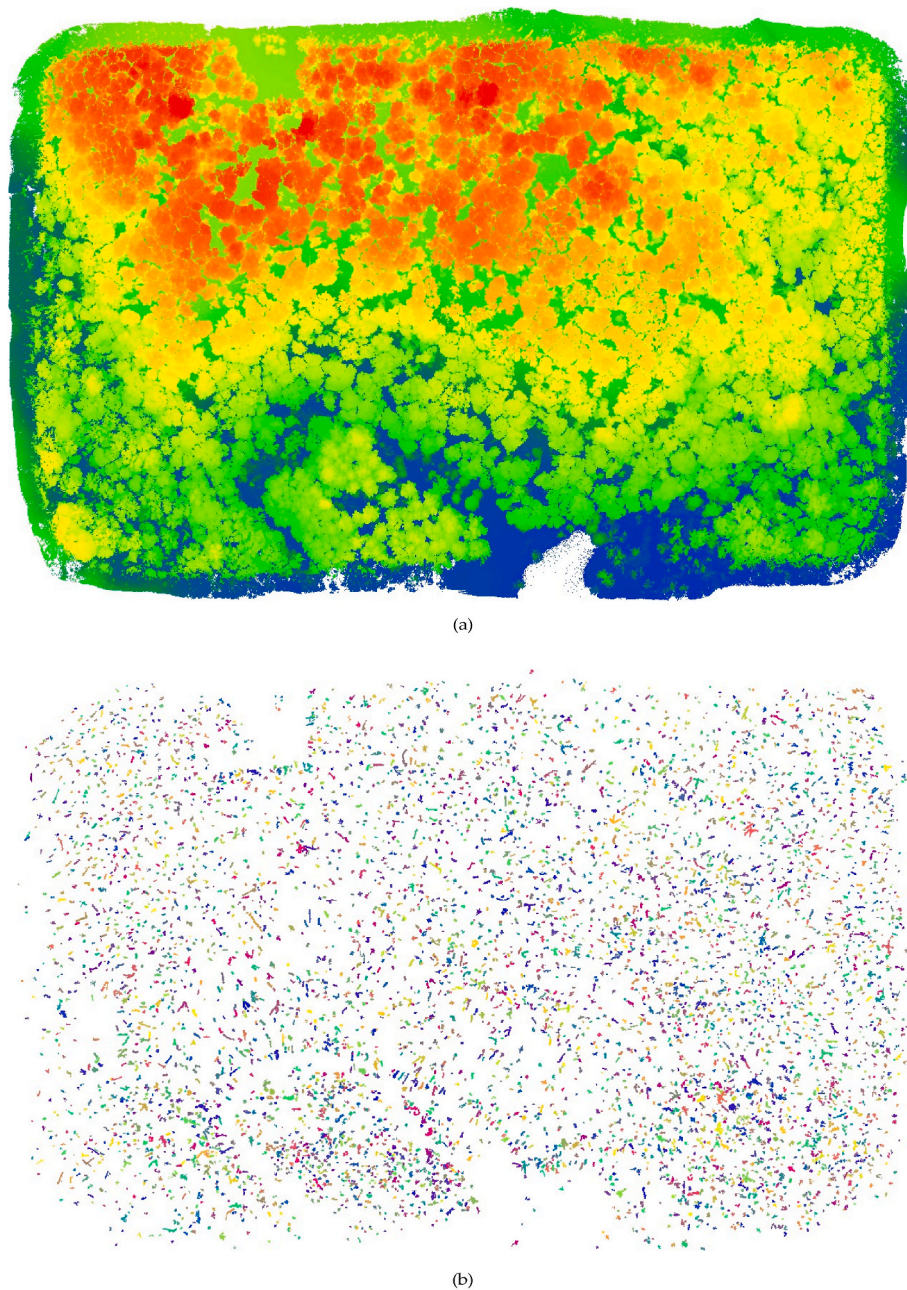


Fig. 7. Illustration of large-scale forest inventory for Plot4B_UAV2 (Bird's eye view). (a) Point cloud colored by height for the entire area of Plot4B_UAV2. (b) Stem mapping result colored by individual tree trunks. The entire Plot4B_UAV2 contains 436 million points within 20 ha. UAV data collection took 60 min and data processing took 20 min. (The growth of trees is not always straight to the air, so some of the individual stems in the stem map in the bird's eye view are a longer shape instead of a dot.)

5.4. Limitations and future directions

The accuracy of DBH measurement is limited by the quality of the LiDAR point cloud. Point density is the key factor that influences the performance of circle fitting. For Plot4D_BP, all cross sections are consisted of complete circles because of the high point density of backpack LiDAR; therefore, all stems are accurately measured regardless of small or large DBH. However, for data from Plot4D_UAV1 and Plot4D_UAV2, DBH measurements are worse than data from backpack LiDAR on small trees. While our framework can achieve high stem mapping accuracy on both UAV LiDAR data, DBH measurements were relatively less precise for trees in the smaller DBH groups (i.e., DBH ranging from 12.7 to 25.4 cm). This is because the UAV LiDAR sensor is far away from the ground, and the forest canopy occludes a part of laser pulses. Point density of stem point clouds from UAV LiDAR, therefore, is lower than those from backpack LiDAR. Although all the data were collected during the leaf-off season which allows for greater penetration of the LiDAR through the canopy, the point density of small trees is not high enough to achieve high accuracy in terms of DBH measurement.

This research demonstrates that our ForestSPG framework is effective in performing semantic segmentation of complex natural forest components. However, our current implementation focuses on relatively basic forest inventory tasks, such as stem mapping and DBH measurement. In the future, we plan to optimize our model to enhance performance in the segmentation of debris and shrubs. Additionally, we aim to explore the effects of changing UAV LiDAR point cloud density on DBH measurement accuracy. We also envision extending our framework to encompass a broader range of forest inventory attributes, including timber volume, biomass estimation, and carbon storage assessment.

6. Conclusion

Forest inventory in a large area has been a challenging problem. In this research, by leveraging flexible and scalable UAV and backpack LiDAR data and deep learning methods, we present a novel framework for large-scale inventory in natural forests. The proposed deep learning method, ForestSPG, has demonstrated the ability to extract various tree components from a massive number of points in complex natural forest environments. Moreover, our experiments have confirmed that the ForestSPG is able to extract robust forest point cloud features that generalize to unseen data from other LiDAR systems that are not included in the training dataset. Based on accurately segmented stem points, stem mapping becomes simple and fast, so that least square adjustment circle fitting can be directly used on high point density LiDAR data. Our framework is able to process 20 ha with hundreds of millions of points input, localize, and measure 5838 trees in 80 min, which is much faster than conventional methods. This research proves that Mobile LiDAR Mapping Systems with modern deep learning methods have the potential to be a practical large-scale natural forest inventory tool for the academic research and real-world forest management.

CRedit authorship contribution statement

Jinyuan Shao: Writing – original draft, Visualization, Validation, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Yi-Chun Lin:** Methodology. **Cameron Wingren:** Writing – original draft, Data curation. **Sang-Yeop Shin:** Writing – original draft, Data curation. **William Fei:** Writing – original draft, Data curation. **Joshua Carpenter:** Writing – original draft, Resources. **Ayman Habib:** Writing – original draft, Conceptualization. **Songlin Fei:** Writing – original draft, Resources, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgement

Funding for this research is partially supported by USDA NIFA # 2023-68012-38992.

Data availability

The data that has been used is confidential.

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